SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

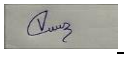
CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 50**
Question Count: 5
Duration : 60 Minutes

Code of Conduct

I Karnesh V K / 192524296 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Equivalent Steps in VMware Workstation Pro

- Library pane showing multiple VMs = multiple instances/regions.
- VM → Snapshot → Take Snapshot, then Snapshot Manager = backup/versioning demonstration.
- Edit → Virtual Network Editor to show NAT/Host-only/Bridged networks = hybrid or multi-cloud connectivity.
- VM → Manage → Clone = replica/auto-scaling demonstration.

Tip: Paste each screenshot directly under the relevant question in this document (Insert → Picture in Word) before final submission, with a one-line caption explaining what it demonstrates.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical application. Analyze performance, security, and manageability, then justify the better choice.

Answer:

A hypervisor is software that creates and manages virtual machines (VMs), allowing multiple operating systems to run on a single physical server. Hypervisors are classified into two types: Type-1 and Type-2.

Type-1 Hypervisor (Bare-Metal Hypervisor):

A Type-1 hypervisor runs directly on the physical hardware without requiring a host operating system. Examples include VMware ESXi, Microsoft Hyper-V, and Xen. It provides high performance because it has fewer software layers between the virtual machines and the hardware. It also offers strong isolation and a smaller attack surface compared with a typical hosted virtualization setup. Centralized management tools help administrators monitor, allocate, and control resources across multiple servers.

Type-2 Hypervisor (Hosted Hypervisor):

A Type-2 hypervisor runs on top of a host operating system such as Windows, Linux, or macOS. Examples include Oracle VirtualBox and VMware Workstation. It is easy to install and suitable for development, testing, and educational purposes. However, its performance may be affected by the host operating system and other running applications. Security also depends on the security of the host operating system.

Comparison:

Criteria	Type-1 Hypervisor	Type-2 Hypervisor
Performance	High performance with low overhead	More overhead due to the host OS
Security	Strong isolation and reduced host OS dependency	Depends partly on the host OS security
Manageability	Supports centralized enterprise management	Generally easier for individual desktop use
Scalability	Suitable for large cloud datacenters	Best for small-scale testing and development
Examples	VMware ESXi, Xen, Hyper-V	VMware Workstation, VirtualBox

Conclusion:

A Type-1 hypervisor is the better choice for a cloud datacenter hosting mission-critical application. It provides better performance, strong workload isolation, efficient resource management, and enterprise-level scalability. These features help maintain application availability, reliability, and security.

2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.

Answer:

Virtualization allows multiple virtual machines to operate on a single physical server while maintaining logical separation between workloads. A cloud provider can use this architecture to host finance, healthcare, and student portal applications on the same hardware without allowing one workload to access another workload's data or resources.

How virtualization supports isolation:

1. Virtual machine isolation: Each VM runs its own operating system and applications. The hypervisor separates memory, CPU resources, and virtual devices assigned to different VMs.
2. Resource allocation: CPU, RAM, and storage can be allocated according to each workload's requirements. Resource limits help prevent one application from consuming all available resources.
3. Network isolation: Virtual networks, VLANs, virtual switches, and firewall rules separate finance, healthcare, and student portal traffic. Access between networks is permitted only when authorized.
4. Storage isolation: Separate virtual disks, access permissions, and encryption help prevent unauthorized access to sensitive information.
5. Access control: Role-based access control (RBAC), multi-factor authentication (MFA), and least-privilege permissions restrict administrative access to the required resources.

Two major attack surfaces:

1. Hypervisor attack surface: The hypervisor manages virtual machines and interacts with physical hardware. A vulnerability in the hypervisor could allow an attacker to escape from one VM and potentially access another VM or the host. This is known as a VM escape risk.
2. Virtual network and management interfaces: Virtual switches, APIs, remote administration interfaces, and management consoles may be targeted through weak passwords, misconfigurations, or unpatched vulnerabilities. An attacker who compromises these components may intercept traffic or modify VM settings.

Security measures:

- Regularly patch and update the hypervisor.
- Separate sensitive workloads using virtual networks and security policies.
- Encrypt sensitive data at rest and in transit.

- Enforce MFA and least-privilege access.
- Monitor logs, network traffic, and unusual VM activity.
- Perform regular vulnerability assessments and security audits.

Conclusion:

Virtualization enables multiple organizations or departments to share physical hardware while maintaining logical isolation. However, secure configuration, hypervisor protection, network segmentation, and continuous monitoring are essential to protect finance, healthcare, and student data.

3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.

Answer:

Virtual machine (VM) host utilization helps administrators identify performance problems in a virtualized environment. In the given scenario, the CPU utilization is 92%, memory utilization is 88%, storage I/O latency is 35 ms, and average VM boot time is 170 seconds. These values indicate that the host may be experiencing resource pressure and performance bottlenecks.

Analysis of the given utilization:

Parameter	Observed value	Analysis
CPU utilization	92%	High CPU usage may cause processing delays and increased VM response time.
Memory utilization	88%	Limited available memory may lead to memory contention and swapping.
Storage I/O latency	35 ms	Elevated storage latency can delay disk reads and writes.
Average VM boot time	170 seconds	Slow boot time may indicate CPU, memory, storage, or startup configuration issues.

Likely bottlenecks:

1. CPU bottleneck:

CPU utilization of 92% indicates that most processing capacity is being used. When several VMs demand CPU resources simultaneously, they may experience scheduling delays. Applications can become slow, and response times may increase.

Corrective actions:

- Allocate CPU resources according to workload requirements.

- Identify and stop unnecessary background processes.
- Balance VMs across multiple physical hosts.
- Upgrade processors or add host capacity when necessary.
- Monitor CPU-ready time to identify VM scheduling delays.

2. Memory bottleneck:

Memory utilization of 88% suggests that the host has limited free RAM. If the memory demand exceeds available capacity, the hypervisor may reclaim memory or rely on swapping, which can reduce application performance.

Corrective actions:

- Identify VMs consuming excessive memory.
- Remove unnecessary memory allocations where appropriate.
- Increase physical RAM if required.
- Migrate VMs to hosts with available memory.
- Monitor memory pressure and swapping activity.

3. Storage I/O bottleneck:

Storage I/O latency of 35 ms may indicate that disk operations are taking longer than expected. Possible causes include overloaded storage devices, excessive simultaneous read/write operations, insufficient storage performance, or network storage congestion.

Corrective actions:

- Monitor disk queues and storage performance.
- Move heavily used VMs to faster storage.
- Use SSD or NVMe storage where appropriate.
- Distribute disk-intensive workloads across storage devices.
- Optimize unnecessary disk operations and investigate storage network congestion.

4. Slow VM boot time:

An average boot time of 170 seconds indicates that virtual machines take nearly three minutes to start. This may be caused by resource contention, slow storage, excessive startup services, or operating system configuration problems.

Corrective actions:

- Reduce unnecessary startup applications and services.
- Investigate storage latency during VM startup.
- Ensure sufficient CPU and memory resources.
- Optimize VM templates and startup configurations.
- Use controlled VM startup scheduling to avoid simultaneous resource demand.

Conclusion:

The primary concerns are high CPU utilization, limited memory availability, elevated storage latency, and slow VM startup. The administrator should first monitor resource usage and identify the processes and VMs responsible for the bottlenecks. Resource optimization, workload balancing, faster storage, and additional hardware capacity can improve VM performance and reduce boot time.

4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.

Answer:

A Software Defined Datacenter (SDDC) is a datacenter architecture in which computing, storage, and networking resources are virtualized and managed primarily through software. It reduces dependence on manually configured hardware and enables centralized, automated resource management.

In a traditional datacenter, administrators often configure physical servers, storage systems, and network devices separately. This can make resource allocation time-consuming and difficult to scale. An SDDC improves agility by automating these operations and allowing resources to be provisioned dynamically according to application requirements.

1. Compute virtualization:

Compute virtualization allows multiple virtual machines to run on a single physical server using a hypervisor. Each VM can run its own operating system and applications.

In an SDDC, administrators can create, modify, clone, or migrate VMs through centralized management tools. Resources such as CPU and memory can be allocated based on workload requirements.

Benefits:

- Faster provisioning of virtual machines.
- Improved utilization of physical servers.
- Easier workload migration and scaling.
- Reduced hardware dependency.
- Better support for disaster recovery.

For example, when an organization experiences increased demand for an application, administrators can provision additional VMs without purchasing and configuring a separate physical server for every instance.

2. Storage virtualization:

Storage virtualization combines physical storage resources into logical storage pools that can be allocated to virtual machines and applications.

Instead of managing each storage device independently, administrators can provision storage

through software-defined policies. Storage can also be expanded or reassigned according to changing business requirements.

Benefits:

- Flexible storage allocation.
- Centralized storage management.
- Improved storage utilization.
- Easier backup and recovery.
- Support for automated storage provisioning.

For example, a database application requiring additional storage can receive capacity from a shared storage pool without requiring extensive manual reconfiguration.

3. Network virtualization:

Network virtualization creates logical networks, switches, routers, and security policies using software. It separates logical network configuration from the underlying physical network infrastructure.

Administrators can create isolated virtual networks for different departments or applications and manage network policies centrally.

Benefits:

- Faster network configuration.
- Improved workload isolation.
- Flexible network segmentation.
- Easier deployment of security policies.
- Simplified support for hybrid and multi-cloud environments.

For example, finance, healthcare, and student applications can operate on logically separated virtual networks even when their VMs share the same physical server infrastructure.

Comparison between traditional datacenter and SDDC:

Feature	Traditional Datacenter	Software Defined Datacenter
Resource management	Often hardware-specific and manual	Software-based and centralized
Compute	Physical servers are provisioned individually	VMs can be created and scaled dynamically
Storage	Devices are often managed separately	Storage is pooled and provisioned logically
Networking	Frequently depends on	Virtual networks and

	manual device configuration	policies can be automated
Scalability	May require lengthy hardware deployment	Resources can be allocated more rapidly
Agility	Slower changes and provisioning	Faster provisioning and configuration
Automation	Limited or dependent on separate tools	Supports policy-driven automation

Conclusion:

SDDC architecture improves datacenter agility by integrating compute, storage, and network virtualization with centralized management and automation. It enables faster resource provisioning, efficient utilization, easier scaling, and more consistent security policies. Therefore, an SDDC is well suited to modern cloud environments where workloads and resource requirements change frequently.

5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Answer:

Multi-cloud federation allows organizations to use services from multiple cloud providers while coordinating identity, access, and resource management. It enables applications and users to access authorized resources across different cloud environments.

However, a multi-cloud federation project may fail when different cloud providers use incompatible identity formats, authentication methods, user identifiers, or access-control policies. These identity mismatches can cause authentication failures, incorrect permission assignments, and security risks.

Causes of identity mismatches:

1. Different identity formats:

Cloud providers may represent users differently. For example, one provider may identify a user using an email address, while another uses a unique user ID. If these identities are not mapped correctly, the same user may be treated as different users across providers.

2. Incompatible authentication mechanisms:

Different cloud platforms may support different authentication protocols or configurations. If the providers cannot establish a compatible trust relationship, users may be unable to access services across clouds.

3. Different authorization policies:

A user may have administrator privileges in one cloud but only read-only access in another. If roles

and permissions are mapped incorrectly, users may receive excessive privileges or be denied legitimate access.

4. Inconsistent identity lifecycle management:

When a user joins, changes roles, or leaves an organization, identity changes must be reflected across all participating cloud providers. Delays in updating accounts can leave unauthorized access active.

Proposed secure identity and privacy design:

1. Implement federated identity management:

Use a trusted Identity Provider (IdP) to authenticate users and establish trust with the participating cloud providers. Federation allows users to access multiple services using a common trusted identity without maintaining separate passwords for every cloud.

Standards such as Security Assertion Markup Language (SAML) and OpenID Connect (OIDC) can support federated authentication, depending on the providers' capabilities.

2. Use consistent identity mapping:

Create a common identity-mapping strategy that links user identities across providers. Use stable, unique identifiers rather than relying only on usernames or email addresses.

For example, a user authenticated by the organization's IdP can be mapped to the corresponding identity in each cloud environment.

3. Apply role-based access control:

Implement Role-Based Access Control (RBAC) and the principle of least privilege. Define common organizational roles and map them to each cloud provider's permissions.

For example, a student may receive read-only access to educational resources, while a cloud administrator receives privileged access only where required.

4. Use secure token-based authentication:

Use signed authentication tokens or assertions to communicate verified identity information. Validate signatures, issuer, audience, expiration, and other required claims. Protect tokens during transmission and prevent replay attacks through appropriate protocol controls.

5. Protect user privacy:

Share only the identity attributes required by the receiving cloud service. Avoid transmitting unnecessary personal information such as complete user profiles when a unique identifier and relevant authorization claims are sufficient.

Use encryption for data in transit and at rest, restrict access to identity logs, and establish clear policies for data retention and sharing between providers.

6. Enforce multi-factor authentication:

Require multi-factor authentication for sensitive applications and privileged accounts. Combine this

with conditional access policies and monitoring to reduce the risk of unauthorized access.

7. Monitor and audit federation activity:

Maintain centralized visibility into authentication events, permission changes, and failed access attempts. Regularly review trust relationships, revoke outdated permissions, and investigate suspicious activity.

Proposed architecture:

The following sequence describes the secure federation process:

1. A user requests access to an application hosted by a cloud provider.
2. The cloud provider redirects the user to the trusted Identity Provider.
3. The Identity Provider authenticates the user and verifies additional security requirements.
4. The Identity Provider issues a signed SAML assertion or OIDC token.
5. The cloud provider validates the token and maps the identity and roles to its local authorization policies.
6. The provider grants access only to permitted resources.
7. Authentication and access events are recorded for auditing and security monitoring.

Conclusion:

Identity mismatches in multi-cloud federation can cause authentication failures, inconsistent authorization, and privacy risks. A secure design should use a trusted Identity Provider, standardized federation protocols, consistent identity mapping, role-based access control, multi-factor authentication, and data minimization. These measures improve interoperability between cloud providers while protecting user identities and sensitive information.

Final submission checklist

- Copy all five answers into your Word document under the corresponding questions.
- Use clear headings and maintain consistent formatting.
- Include the comparison tables where appropriate.
- Add relevant VMware Workstation screenshots if your instructor requires them. The uploaded assessment document suggests demonstrating multiple VMs, snapshots, virtual networks, and VM cloning.
- Add your name, registration number, and signature in the assessment's code-of-conduct section.